

LAB-01 ASSIGNMENT

Building a Spring Boot Books REST API

Enterprise Web Application Development | Kabul University

Course: Enterprise Web Application Development

Lab Topic: Spring Boot REST API Fundamentals

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1. Lab Objectives

- Implement a PUT endpoint to update an existing resource.
- Implement a DELETE endpoint to remove an existing resource.
- Use @PutMapping, @DeleteMapping, @PathVariable, and @RequestBody.
- Return appropriate HTTP status codes such as 200 OK, 204 No Content, and 404 Not Found.
- Test PUT and DELETE requests using Postman.

2. Get all books

```
[
  {
    "id": 2,
    "title": "Modern Web Development",
    "author": "Sara Ahmad",
    "isbn": null,
    "publishedYear": 0,
    "category": null
  },
  {
    "id": 3,
    "title": "Designing Data-Intensive Applications",
    "author": "Martin Kleppmann",
    "isbn": "9781449373320",
    "publishedYear": 2017,
    "category": "Distributed Systems"
  },
  {
    "id": 4,
    "title": "Spring in Action",
    "author": "Craig Walls",
    "isbn": "9781617297571",
    "publishedYear": 2022,
    "category": "Spring"
  },
  {
    "id": 5,
    "title": "Computer Networks",
    "author": "Andrew S. Tanenbaum",
    "isbn": "9780132126953",
    "publishedYear": 2010,
    "category": "Networking"
  }
]
```

3. PUT updates an existing book

PUT Get data

Overview

+

My Collection / Get data

PUT

▼

http://localhost:8080/api/v1/books/2

Docs

Params

Authorization

Headers (8)

Body

Scripts

Settings

☐ none

☐ form-data

☐ x-www-form-urlencoded

☒ raw

☐ binary

☐ GraphQL

JSON ▼

```
1  {
2    "title": "Modern Web Development",
3    "author": "Sara Ahmad",
4    "availableCopies": 6
5  }
```

Body

Cookies

Headers (5)

Test Results (1/1)

🔄

200 OK

{ } JSON ▼

▶ Preview

🖼 Visualization ▼

```
1  {
2    "id": 2,
3    "title": "Modern Web Development",
4    "author": "Sara Ahmad",
5    "isbn": null,
6    "publishedYear": 0,
7    "category": null
8  }
```

4.PUT returns 404 for a missing book

PUT Get data

Overview

HTTP

My Collection / Get data

PUT

http://localhost:8080/api/v1/books/21

Docs

Params

Authorization

Headers (8)

Body

Scripts

Settings

none

form-data

x-www-form-urlencoded

raw

binary

GraphQL

JSON

1 {

2 "title": "Modern Web Development",

3 "author": "Sara Ahmad",

4 "availableCopies": 6

5 }

Body

Cookies

Headers (4)

Test Results

404 Not Found

Raw

Preview

Debug with AI

1

5.DELETE removes an existing book

The screenshot displays a REST client interface with a dark theme. At the top, there are tabs for 'DEL Get data' (active), 'Overview', and a '+' icon. Below the tabs, the breadcrumb 'My Collection / Get data' is shown. The main area is divided into two sections: the top section for the request and the bottom section for the response.

Request Section:

- Method:** DELETE
- URL:** http://localhost:8080/api/v1/books/2
- Body Type:** raw (selected), with other options like none, form-data, x-www-form-urlencoded, binary, and GraphQL.
- Body Content:**

```
1  {  
2  "title": "Modern Web Development",  
3  "author": "Sara Ahmad",  
4  "availableCopies": 6  
5  }
```

Response Section:

- Status:** 204 No Content
- Body Type:** Raw (selected), with other options like Preview and Visualize.
- Body Content:**

```
1
```

6.DELETE returns 404 for a missing book

The screenshot shows a REST client interface with the following components:

- Top Bar:** Includes tabs for "DEL Get data" (active), "Overview", and a "+" icon.
- Breadcrumb:** "My Collection / Get data".
- Request Section:**
 - Method:** "DELETE" (highlighted with a blue box).
 - URL:** "http://localhost:8080/api/v1/books/21" (highlighted with a blue box).
 - Docs:** A menu icon.
 - Params:** A tab.
 - Authorization:** A tab.
 - Headers (8):** A tab.
 - Body:** A tab (active), showing a JSON object:

```
1 {  
2   "title": "Modern Web Development",  
3   "author": "Sara Ahmad",  
4   "availableCopies": 6  
5 }
```
 - Scripts:** A tab with a red diamond icon.
 - Settings:** A tab.
- Response Section:**
 - Body:** A tab (active), showing a "404 Not Found" error.
 - Cookies:** A tab.
 - Headers (4):** A tab.
 - Test Results:** A tab with a red dot icon.
 - Raw:** A dropdown menu (active), showing "1".
 - Preview:** A button.
 - Debug with AI:** A button.